RWANDA

COUNTRY REPORT ON GENOME EDITING LANDSCAPE ANALYSIS ON AGRICULTURE

MAY 2026

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LIST OF ABBREVIATIONS AND ACRONYMS

AATF	African Agricultural Technology Foundation
ABNE	African Biosafety Network of Expertise
ABS	Access and Benefit-Sharing
AfDB	African Development Bank
AfPBA	African Plant Breeding Academy
AOCC	African Orphan Crops Consortium
APAARI	Asia-Pacific Association of Agricultural Research Institution
APCoAB	Asia-Pacific Consortium on Agricultural Biotechnology (APCoAB)
ARIPO	African Regional Intellectual Property Organization
AUDA-NEPAD	African Union Development Agency- New Partnership for Africa's Development .
BecA-ILRI	Biosciences eastern and central Africa–International Livestock Research Institute
BIOCAP	Biotech Capacity building
BIRAC	Biotechnology Industry Research Assistance Council
BSc	Bachelor of Science
CL	Containment Level
CAADP	Comprehensive Africa Agriculture Development Programme
CAVM	College of Agriculture, Animal Sciences and Veterinary Medicine
CBD	Convention on Biological Diversity
CBSD	Cassava Brown Streak Disease
COMESA	Common Market for Eastern and Southern Africa
GDP	Gross Domestic Product
CGIAR	the Consultative Group on International Agricultural Research
CIAT	Alliance of Bioversity International and CIAT (International Centre for Tropical Agriculture)
CIP	International Potato Centre

CIMMYT	International Maize and Wheat Improvement Centre
CIRAD	French Agricultural Research Centre for International Development
CMD	Cassava Mosaic Disease
CSAS	Climate Smart Agriculture Strategy
CST	College of Science and Technology
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
CRISPR/Cas9	Clustered Regularly Interspaced Short Palindromic Repeats/CRISPR-associated protein 9
EAC	East African Community.
ECOLEX	A web-based information service that synergises information on environmental law collected by the three partner organisations.
EPA	Environmental Protection Agency
ETG	Export Trading Group
EU	European Union
FAO	Food and Agriculture Organisation of the United Nations.
FNIH	Foundation for the National Institutes of Health.
GEd	Genome Editing
GMO	Genetically Modified Organism
GIZ	German Agency for International Cooperation
HoReCo	Horticulture in Reality Cooperation Ltd.
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics.
IGI	Innovative Genomics Institute
IITA	International Institute of Tropical Agriculture
IIPCC	International Intellectual Property Commercialisation Council
ILRI	International Livestock Research Institute
INES	Institut d'Enseignement Superieur /Institute of Applied Sciences_Ruhengeri
IPR	Intellectual Property Right
IRRI	International Rice Research Institute.

LMOs	Living Modified Organisms
MINAGRI	Ministry of Agriculture and Animal Resources
MSU	Michigan State University
NAIP	National Agriculture Investment Plan
NAP	National Agricultural Policy
NBC	National Biosafety Committee
NBT	New Breeding Techniques.
NRL	National Reference Laboratory
NISR	National Institute of Statistics of Rwanda
NST	National Strategy for Transformation
ODK	Online data kit
OAPI	African Intellectual Property Organisation
OFAB	Open Forum on Agricultural Biotechnology in Africa
OHADA	Organisation for the Harmonisation of Law in Africa
PBR	Plant Breeders Rights
PCR	Polymerase Chain Reaction
PCT	Patent Cooperation
PPP	Public-Private Partnerships.
PSTA4	Fourth Strategic plan for Agriculture Transformation
R&D	Research and Development
RAB	Rwanda Agriculture & Animal Resources Development Board
RBC	Rwanda Biomedical
RDB	Rwanda Development Board
PSTA V	Fifth Strategic Plan for Agricultural Transformation
REMA	Rwanda Environment Management Authority
RFA	Rwanda Forest Authority
RICA	Rwanda Inspectorate, Competition and Consumer Protection Authority
RSB	Rwanda Standards Board.

RP	Rwanda Polytechnic
RSB	Rwanda Standards Board
SAIP	Sustainable Agriculture Intensification for Improved Livelihoods, Food Security and Nutrition Project.
SPC	Sanitary and Phytosanitary Controls
TAAT	Technologies for African Agricultural Transformation
TELA	Tela Maize Project
TReND	Teaching and Research in (Neuro) science for Development
TRIPS	The Agreement on Trade-Related Aspects of Intellectual Property Rights
TWAS	Third World Academy of Science
ULK	Kigali Independent University
UNEP	United Nations Environmental Program
UNFCCC	United Nations Framework Convention on Climate Change
UR	University of Rwanda
UR-CHG	University of Rwanda- College of Science & Technology; Centre for Human Genetics.
USAID	United States Agency for International Development
UTAB	University of Technology and Arts of Byumba
WAITRO	World Association of Industrial and Technological Research Organisations
WIPO	The World Intellectual Property Organisation

EXECUTIVE SUMMARY

Background: Genome editing (GEd) in Rwanda is at an early but promising stage. While large-scale deployment has not yet occurred, the country has laid important policy foundations, developed institutional capacity, and aligned national priorities that support the adoption of this technology. With sustained investment in research, regulation, and regional collaboration, Rwanda is well positioned to leverage GEd to advance its food security and climate resilience objectives. Progress toward genome editing and the potential uptake of genome-edited products is evident, with multiple institutions engaged in the regulation and evaluation of genetically engineered organisms in accordance with the national biosafety framework. GEd technology holds significant potential to address pressing challenges related to climate change, pests and diseases, nutrition, and crop productivity. However, increased funding and long-term investment are required to strengthen laboratory infrastructure and equipment, build technical capacity, and enhance public awareness.

The African Union Development Agency–New Partnership for Africa's Development (AUDA-NEPAD) commissioned a Genome Editing Landscape Analysis of Rwanda to provide an in-depth assessment of the country's readiness for genome editing research, product development, and commercialisation. The assignment sought to evaluate existing policies, infrastructure, institutional arrangements, and technical capabilities.

Data for the assignment were collected from both secondary and primary sources. Secondary data included published literature and institutional databases, including official websites of relevant stakeholders. Primary data were obtained through interviews conducted using the Online Data Collection Kit (ODK), structured surveys, and email-based questionnaires. Data from both sources were analysed, synthesized, and organized across thematic areas, including participation in multilateral food and environmental agreements, regulatory and policy frameworks, socio-economic considerations in decision-making, genome editing projects and programs, human and institutional capacity, research and academic institutions, training and professional development, infrastructure and equipment, priority crops for genome editing, intellectual property rights, and funding sources.

Key findings

Genetic modification (GM) technology including GEd is at a nascent stage in Rwanda. The Rwanda Environment Management Authority (REMA), under the Ministry of Environment, is responsible for modern biotechnology and biosafety management. REMA exercises its mandate in consultation with six key regulatory agencies: the Rwanda Food and Drugs Authority, Rwanda Standards Board (RSB), Rwanda Agriculture and Animal Resources

Development Board, Rwanda Inspectorate, Competition and Consumer Protection Authority, Rwanda Development Board (RDB), and Rwanda Forestry Authority (RFA).

The GEd technology in Rwanda is governed by the Law Governing Biosafety (Law No. 025/2024 of 16/02/2024), and the national regulatory framework for GEd is still under development. Universities, research institutions, and private companies operating in Rwanda possess varying but complementary strengths and play important roles in GEd-related research, training, and potential implementation. Strategic partnerships among these institutions, alongside support from funding organisations, are essential for advancing innovation and the responsible use of new breeding tools. Some institutions already have the basic infrastructure and human resource to conduct genome editing research.

GEd technology demonstrates strong potential for improving yield, stress tolerance, and nutritional traits in key staple and traditional crops, including rice, beans, maize, banana, cassava, sugarcane, and Irish potato. Currently, Rwanda has no GEd projects. However, there are ongoing research on GMOs led by government research institutes and universitiess, which are focusing on cassava, maize, and Irish potato targeting diseases, pests, and climate-related stresses. Private-sector involvement remains limited and is confined to experimental or pilot-scale field trials. Funding for GMOs research is derived from government line ministries and external investors through PPPs. Intellectual property rights in Rwanda’s agricultural sector are protected through legal instruments covering patents, copyrights, and plant breeders’ rights.

Policy implications and recommended actions

Genome editing presents a significant opportunity to strengthen food security, enhance climate resilience, and accelerate agricultural modernisation in Rwanda. Realizing this potential will require coherent and enabling policies, strong biosafety governance, sustained public engagement, and regional collaboration. If effectively managed, genome editing could position Rwanda as a leader in agricultural biotechnology in Africa, in alignment with the country’s Vision 2050 goals for inclusive prosperity and sustainable development.

Conclusion

Rwanda currently has no GEd projects. However, with initiation of R&D on GMOs, a great interest is gaining momentum to exploit the potential of GEd technology to improve crop yields, disease resistance, and climate resilience in Rwanda. With clear and enabling policies and guidelines, targeted investments in human resources and infrastructure, and strong collaboration among stakeholders, GEd technology can contribute to climate-resilient, high-yielding crops, improved livestock and fisheries, and a strengthened forestry sector—thereby advancing Rwanda’s national food security objectives.

1 INTRODUCTION

1.1. Description of Rwanda's Agricultural Landscape

Agriculture is the backbone of Rwanda's economy, employing approximately 70% of the labour force and contributing about 25-30% of Gross Domestic Product (GDP). The sector is central to rural livelihoods, poverty reduction, and national food security. Rwanda's long-term development vision, articulated in Vision 2050 and operationalized through the National Strategy for Transformation (NST1), identifies agriculture as a priority sector for modernisation, productivity growth, and inclusive economic development.

Rwanda's agricultural landscape is dominated by smallholder farming systems, with a strong focus on staple food crop production alongside export-oriented crops such as coffee and tea. The sector faces persistent challenges, including land scarcity, soil degradation, climate variability, and increasing pressure on natural resources. In response, government policies and innovation-driven interventions seek to transform agriculture into a modern, productive, and climate-resilient system capable of ensuring food security, improving livelihoods, and supporting sustained economic growth.

1.2. National Frameworks and Investments.

Rwanda's national development and agricultural policy frameworks provide a solid foundation for integrating genome editing into the country's broader agricultural transformation agenda. Key instruments, including the National Agricultural Policy (NAP), the Strategic Plan for Agriculture Transformation Phase IV (Ministry of Agriculture and Animal Resources, 2018), and the National Agriculture Investment Plan (NAIP) (Ministry of Agriculture and Animal Resources, 2009), emphasize agricultural modernisation, climate resilience, and food and nutrition security through the application of science, technology, and innovation.

Within this policy context, GEd aligns closely with Rwanda's strategic priorities for promoting climate-smart, high-yielding, and nutrition-sensitive agricultural systems. The technology offers potential pathways to address productivity constraints, enhance resilience to climate stresses, and improve the nutritional quality of key crops, thereby supporting national development objectives.

1.3. Engagement in Regional Frameworks.

Rwanda's biosafety and biotechnology policy frameworks are informed by its commitments under the Cartagena Protocol on Biosafety and reinforced through regional and continental harmonisation efforts. Engagement within the East African Community (EAC) and the African Union (AU) supports the responsible and safe adoption of emerging biotechnologies while facilitating regional cooperation and trade. By linking national policies with regional frameworks, Rwanda is fostering an enabling environment in which genome editing can be applied to address persistent challenges—such as crop diseases, land constraints, and nutritional deficiencies—while safeguarding public trust and ensuring benefits reach smallholder farmers.

At the continental level, Rwanda has actively engaged with the African Union Development Agency–New Partnership for Africa's Development (AUDA-NEPAD) through capacity-building initiatives and technical dialogues on agricultural biotechnology. The country has also expressed interest in the African Union's 2022 continental guidelines on genome editing, which encourage member states to adopt enabling, science-based policies while ensuring biosafety, ethical oversight, and public accountability. Within the EAC, Rwanda has participated in efforts to harmonize biosafety regulations to support cross-border collaboration, knowledge exchange, and regional trade in agricultural biotechnology products. In addition, Rwanda aligns with the principles of the Comprehensive Africa Agriculture Development Programme (CAADP) (African Union Commission, 2003), which recognizes science, technology, and innovation as key drivers of agricultural transformation and food system resilience.

1.4. Challenges and Opportunities: Justification of GEd Assessment based on Current Issues Identified in Rwanda.

1.4.1. Challenges Justifying GEd Assessment.

To realize the potential of genome editing, Rwanda will require addressing challenges related to regulatory clarity, limited technical and institutional capacity, public awareness participation and perception, and inclusive policy processes. Effectively navigating these challenges will be critical to unlocking the full benefits of genome editing for sustainable and equitable agricultural development.

1.4.2. Opportunities Supporting GEd Investment.

Genome editing presents Rwanda with significant opportunities to strengthen food security, enhance climate resilience, and advance its innovation-led development agenda. Focus on technology-based improvement of staple and traditional crops on yields, production and resistance traits offers a useful opportunity for harnessing GEd in Rwanda.

1.5. Objectives of the Landscape Analysis.

This landscape analysis aims to provide an evidence-based description and analysis of the status of modern biotechnology and GEd in Rwanda, with a focus on identifying key trends, enabling and constraining factors, and priority areas for policy and investment attention. Specifically, the assessment sought to:

- Identify emerging needs that genome editing can rapidly address, particularly in agricultural productivity, reduction of post-harvest losses, climate adaptation, food and nutrition security, and the promotion of diversified and healthy diets; and
- Identify staple and traditional crops, within both national and broader African regional contexts, which have the potential to improve livelihoods through enhanced food security, nutrition, climate resilience, and sustainable productivity.

2 APPROACH/METHODOLOGY

Secondary data (literature review) for Rwanda were collected from published literature and institutional databases, including official websites of relevant organisations. Primary data were gathered by a team of trained enumerators who visited key institutions and conducted face-to-face interviews with identified stakeholders. In exceptional cases, online links were used to administer questionnaires remotely. Primary data collected through live interviews were captured using the Online Data Collection Kit (ODK) and securely stored on the Africa Harvest server.

Data from both sources - primary and secondary - were subsequently analysed, synthesized, and organized into detailed thematic narratives covering the following areas:

2.1. Regulatory and Policy Frameworks for Biotechnology and Genome Editing

Information on Rwanda's biotechnology and GEd regulatory and policy frameworks was obtained from both secondary and primary sources. These data were retrieved, compiled, synthesized, and presented in tabular formats to provide an overview of the functionality of the existing framework and Rwanda's level of preparedness to adopt and implement genome editing technologies.

2.2. Genome Editing Projects

Data on biotechnology, and particularly genome editing projects, were collected for crops, livestock, fisheries, and forestry sectors. Information on target traits, key stakeholders and partnerships, and funding sources was documented through both secondary and primary data collection. The synthesized and analysed data were used to:

i. Identify emerging needs that genome editing can address in terms of economic, social, and environmental or climate benefits; and
ii. Assess the status of existing human and infrastructural capacity for genome editing technologies in Rwanda.

2.3. Traditional and Staple Crops for Genome Editing Improvement

Data on genome editing projects, target crops, and associated traits were further disaggregated and categorized to identify those with the highest potential for socio-economic impact at the national level. Prioritisation considered the likelihood of successful development and adoption, national acceptance, resource requirements, and scalability of genome editing interventions.

2.4. Institutional Capacity (Human Capital, Infrastructure and Equipment)

During primary data collection, respondents were asked detailed questions on existing institutional capacities, including human resources, laboratory and field infrastructure, and equipment available for genome editing research and development, commercialisation, and scaling. Data were aggregated to generate institutional profiles reflecting current human capital and infrastructure capacity for genome editing in Rwanda.

2.5. Stakeholder Mapping

A targeted sampling approach was employed, focusing on individuals with demonstrated knowledge of, and active engagement in, modern agricultural biotechnology and genome editing across regulation, policy, research and development, and commercialisation. Key stakeholders were identified through:

i. Published literature and scientific outputs.
ii. Institutional databases and official websites
iii. Referrals from institutional leadership or direct knowledge of the country principal investigator.

Stakeholders were categorized into five groups, as reflected in the data collection tools: regulatory agencies, research organisations and institutions, universities, private sector and industry, and government departments, ministries, and policymakers.

2.6. Database Systems and Data Management

Project consortium members and sponsors held technical backstopping meetings to design and refine appropriate data collection tools and digital platforms to support primary data collection. Questionnaires were customized for each stakeholder category—regulatory bodies, research institutions, universities, private sector entities, and government agencies—to generate structured datasets assessing Rwanda's readiness, capabilities, and gaps in adopting and scaling genome editing technologies. All data collection tools and platforms were pre-tested prior to deployment.

Both methodological and data triangulation approaches were employed to reduce bias and enhance the validity and reliability of findings. All collected data were encrypted and stored on a secure, centralized server to ensure participant confidentiality and compliance with ethical standards.

2.7. Data synthesis and Data analysis

Data collected was synthesized and packaged into appropriate information, and where applicable, tables were generated. Variables investigated included, but were not limited to:

i. Signatory to multinational food and environmental agreements;
ii. Regulatory framework components;
iii. Applications received and approved (field testing, registration, and commercialisation);
iv. R&D institutions, personnel, and laboratory/field facilities;
v. Projects;
vi. Crops and livestock species; and
vii. Traits under GEd development.

2.8. Interactive Mapping

An interactive map, modelled on the agenda 2063 dashboard, was developed to support visualisation and exploration of the compiled dataset. The platform enables users to view genome editing projects, crops, and traits spatially and is accessible through the Genome Africa portal (https://genome.africa/).

3. RESULTS

3.1 Membership in Multilateral Environment Treaties and Agreements

Rwanda is a member of the Codex Alimentarius Commission since 1988. The Rwanda Standards Board hosts the National Codex Committee, which provides the institutional platform for the application of Codex standards and guidelines. Consequently, Codex Alimentarius standards serve as the primary international reference for food and feed safety assessment and regulatory decision-making for products derived from genome editing. This alignment is critical for ensuring food safety, facilitating international trade, and promoting regulatory harmonisation. In addition, Rwanda is a Party to the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilisation. Where genome editing activities involve the use of Rwanda's genetic resources or associated traditional knowledge, Access and Benefit-Sharing (ABS) requirements apply. The Rwanda Environment Management Authority (REMA), in collaboration with relevant sector ministries and agencies, coordinates the domestic implementation of ABS provisions under the Nagoya Protocol. Rwanda is also a party to the Convention on Biological Diversity (CBD) and its Cartagena Protocol on Biosafety (see Table 1). By ratifying these treaties and agreements, Rwanda commits to global efforts in areas such as food safety, climate action, biodiversity conservation, and environmental protection, and integrates these obligations into its national food safety and environmental management frameworks.

Table 1: Status of Rwanda's participation in key multilateral environmental agreements

Multilateral Environmental Agreements (MEAs) / Treaties	**Date of Ratification / Accession**	**Notes / Source**
Codex Alimentarius Commission is a joint body of the Food and Agriculture Organization (FAO) and the World Health Organization established to develop international food standards, guidelines, and codes of practice. Critical for risk assessment of food developed through genome editing	Member 1988	https://www.fao.org/fao-who-codexalimentarius/news-and-events/news-details/ru/c/1754844/
Convention on Biological Diversity (CBD)	Signed 1992, Ratified – 27 August 1996	Verified via CBD national profile (Convention on Biological Diversity)

Cartagena Protocol on Biosafety	Ratified – 20 October 2004	CBD profile confirms (Convention on Biological Diversity)
Nagoya Protocol on Access and Benefit-sharing	Singed 2010 Ratified – March 2012	As stated in CBD profile (Convention on Biological Diversity)
UN Framework Convention on Climate Change (UNFCCC)	Signed 1992; Ratified 1998	Confirmed via REMA MEA listing (rema.gov.rw, Index Mundi)
Paris Agreement	Ratified/Party – around October–November 2016	Reported a ratification in 2016 (lands.rw)
Kyoto Protocol	September 2003	Included in Rwanda's REMA list (rema.gov.rw)

3.2 Policies that Advance or Hinder Adoption of Modern Biotech and GEd

Rwanda has three (3) main policies that support the application of modern agricultural biotechnology including GEd. These policies, which emphasize modern technology's central role in enhancing agricultural productivity are:

- **The fifth strategic plan for agricultural transformation (PSTA V) (2024-2029)** explicitly embeds the adoption of modern agricultural biotechnology to meet national economic goals. The strategy essentially targets a 50% increase in agricultural productivity by integrating advanced sciences including genome editing and genetic transformation to protect food security and build climate resilience.
- **Second national strategy for transformation (NST2) (2024-2029)** Rwanda aligns its agricultural biotechnology goals directly with its highest-level national developmental agenda (the NST and Vision 2050) through the Second national strategy for transformation (NST). The strategy champions tech-driven, high-yield agriculture to transition the country from subsistence farming to a knowledge-based, commercial agrarian economy.
- **The Climate Smart Agriculture Strategy (CSA) (2025)** is led by MINAGRI and encourages resilient crop systems and modern breeding technologies including development and deployment of drought-resistant crops, precision farming, improved crop genetics.

3.3 National Regulatory Framework

3.3.1 Biosafety Act/Law, Biosafety Regulations/Decrees and GEd Guidelines

Rwanda's Biosafety Law No. 025/2024 adopts a technology-neutral approach and regulates LMOs developed through modern biotechnology. Genome-edited organisms are assessed on a case-by-case basis to determine whether they meet the statutory definitions and regulatory thresholds that trigger biosafety oversight. Where such criteria are met, genome-edited products are subject to REMA's permitting and risk-assessment processes, with scientific input from the NBC. Regulatory oversight of GEd products extends beyond biosafety to include downstream and sector-specific mandates. Food and feed safety assessments, labelling, and standards are overseen by the Rwanda Food and Drugs Authority and the Rwanda Standards Board (RSB), in accordance with national legislation and Codex Alimentarius principles. Sanitary and phytosanitary controls (SPC), including the regulation of seeds and planting materials and border inspections, are implemented by the Rwanda Inspectorate, Competition and Consumer Protection Authority (RICA), in coordination with REMA's biosafety approvals.

Overall, Rwanda has established a multi-institutional regulatory system with clearly defined mandates for biosafety, food and feed safety, standards, and SPC. These institutions operate under enabling legal and policy instruments that collectively govern the regulation of LMOs and genome-edited products across the research-to-commercialisation continuum (Table 2).

3.3.2 Regulatory Agencies

The Rwanda Environment Management Authority (REMA) is the designated Competent National Authority for biosafety under the Biosafety Law No. 025/2024. REMA is responsible for receiving, reviewing, and making regulatory decisions on applications involving Living Modified Organisms (LMOs) and products of modern biotechnology. Technical review and risk assessment are undertaken with advice from the National Biosafety Committee (NBC).

Within this framework, REMA constitutes the principal regulatory institution overseeing genome-edited organisms that fall within the scope of the biosafety law, including research, confined trials, environmental release, commercialisation, and transboundary movement, where applicable.

NB. Rwanda does not have a GEd guideline.

Table 2: Regulatory and institutional landscape for GEd in Rwanda

Institutions	Mandate / Relevance to GEd	Regulatory instruments	Date of enactment	Coverage / scope	Reference
REMA – Competent National Authority	Leads biosafety regulation; receives and decides on permit applications for activities involving LMOs/modern biotech (including genome-edited materials where covered); keeps national registry; coordinates risk assessment	Law No. 025/2024 governing biosafety (establishes CNA, Registrar, National Biosafety Committee)	Gazetted 21 Feb 2024 (law dated 16 Feb 2024)	Import, export, transit, research, contained/confined use, environmental release and placing on the market; labelling; institutional arrangements	(BWC Implementation, USDA Apps, Rwanda Environment Management Authority)
National Biosafety Committee (NBC)	Scientific/technical advisory body to REMA; undertakes/oversees risk assessment and advises on permits	Established by Law No. 025/2024; earlier composition and roles outlined in Rwanda's biosafety strategy	21 Feb 2024 (law); strategy 2019	Case-by-case scientific review across sectors; multi-agency membership (incl. agriculture, health, standards, customs, etc.)	(BWC Implementation, Rwanda Environment Management Authority)
Ministry of Environment (policy); REMA organic law	Policy oversight for environment; REMA's mission and powers anchored in its establishing law	Law No. 63/2013 determining REMA's mission, organisation and functioning	14 Oct 2013	Establishes REMA's legal personality and powers for environmental regulation and coordination (under which biosafety sits)	(RwandaLII, climate-laws.org)
RAB	National agricultural R&D, breeding and technology development; conducts trials/research that may use genome editing under biosafety permits	Law No. 14/2017 establishing RAB; Presidential Order governing RAB (2022); RAB Strategic Plan 2020–2024	01 May 2017 (law); 09 Dec 2022 (PO)	Agricultural research, extension and innovation; implements regulated trials and technology deployment in coordination with REMA	(RwandaLII, rlrc.gov.rw, FAOLEX Database)

Rwanda FDA	Protects public health through regulation of foods, feeds, human/veterinary medicines and biologicals; relevant for safety/labelling of GM/GE-derived foods placed on the market	Law No. 003/2018 establishing Rwanda FDA; FDA guidance and overview	09 Feb 2018	Food and feed regulation, market surveillance, compliance and labelling within its remit; coordination with REMA on GM products	(RwandaLII, rwandafda.prod.risa.rw, FAOLEX Database)
Rwanda Standards Board (RSB)	National standards body; develops/adopts standards and supports conformity assessment; relevant to testing/labelling standards for biotech products	Law No. 50/2013 establishing RSB; official roles as national standards body	28 Jun 2013	Standards, technical regulations support, metrology and certification that may apply to GM/GE products and labelling	(minicom.gov.rw, ISO, Trade.gov)
Rwanda Inspectorate, Competition and Consumer Protection Authority (RICA)	SPS authority and inspectorate; seed/plant product quality control, quarantine and border inspections; relevant for GM seed/plant consignments alongside REMA permits	Law No. 31/2017 establishing RICA; mandate pages and SPS materials	18 Aug 2017	Quality and standards conformity inspections for agro-inputs, plants/plant products; coordinates SPS controls and serves as NPPO	(RwandaLII, rica.gov.rw, standardsfacility.org)

3.4 Socio-Economic Considerations for Decision-Making on GEd Technologies

Socio-economic considerations are central to policy and regulatory decision-making on genome editing in Rwanda, where agriculture remains the primary source of livelihoods for most of the population. Ensuring equitable access for smallholder farmers is critical so that productivity gains—such as higher yields, disease resistance, and drought tolerance—do not disproportionately benefit large-scale or commercial producers.

Decision-makers must balance the high upfront costs associated with research, regulatory oversight, and capacity development against longer-term economic benefits, including reduced reliance on external inputs, lower post-harvest losses, improved nutritional outcomes through biofortified crops, and enhanced national food security. Consumer acceptance, shaped by cultural values, trust in institutions, and perceptions of safety, will directly influence market uptake and scalability. At the same time, regional and international trade considerations require policy coherence and regulatory harmonisation to avoid potential market barriers.

Attention to equity, gender inclusion, youth participation, and rural employment is essential to ensure that genome editing contributes to inclusive growth rather than exacerbating socio-economic inequalities. Finally, integrating GEd innovation with biodiversity conservation goals and environmental stewardship will be critical in determining whether the technology supports sustainable agricultural transformation in the long term.

3.5 Factors that Hinder or Advance Application of Biotech and GEd in Rwanda

Several factors hinder or advance the application of modern biotechnology including GEd in Rwanda. These factors are political, geopolitical, ssocial, human, culture, and traditions. Socially, trusted farmer cooperatives, extension systems, and community networks provide important platforms for information sharing and collective decision-making and could accelerate awareness and acceptance of GEd technologies. Conversely, misinformation, limited understanding of biotechnology, and fears surrounding “tampered” or externally controlled seeds may slow adoption if not proactively addressed through transparent communication and engagement. Human capital factors particularly education, technical training, and access to resources will influence the extent to which smallholder farmers, researchers, and institutions can benefit from genome editing. While Rwanda has made substantial investments in agricultural capacity building, gaps in specialized skills, infrastructure, and equitable access could reinforce existing disparities if GEd innovations are not deliberately inclusive. Cultural considerations are equally significant. Adoption of GEd technologies is likely to be more acceptable when aligned with Rwanda's staple crops, food preferences, and dietary traditions. Perceptions of "unnaturalness," ethical concerns, or religious objections may generate resistance in some communities, underscoring the importance of culturally sensitive engagement strategies. Traditional practices such as seed saving, farmer-to-farmer exchange, and collective governance of agricultural resources may initially pose challenges if GEd crops are perceived as undermining local autonomy. However, when integrated with traditional knowledge systems and participatory, farmer-led innovation approaches, these same traditions could serve as strong enablers of responsible and locally appropriate adoption.

Political factors that advance Biotech & GEd include strong government commitment to innovation; biotechnology is included in national agricultural transformation (PSTA4, NST1); supportive institutions (MINAGRI, RAB, RICA) and a biosafety framework exists while those that hinder include strict biosafety approval systems; cautious regulatory pathway for GM/GEd crops and dependence on donor-driven biotech projects. **Geopolitical** factors that advance Biotech & GEd include strong partnerships with CGIAR, Gates Foundation, EU, USA universities; regional leadership ambition in agri-biotech and access to global funding while those that hinder include dependence on international partners for expertise and funding; limited global biotech influence and technology import reliance. **Social** factors that advance Biotech & GEd include high trust in government-led agricultural innovation; strong focus on food security and growing acceptance of science-based agriculture while those that hinder include limited public understanding of genome editing vs GMO; misinformation risk and rural adoption barriers. **Human capital** factors that advance Biotech & GEd include an active biotechnology training program (University of Rwanda, BIOCAP); growing pool of molecular biology graduates and diaspora collaborations while those that hinder include shortage of CRISPR/genome-editing specialists; limited bioinformatics capacity and brain drain of advanced researchers. **Cultural** factors that advance Biotech & GEd include, agriculture is seen as national priority; there is openness to improved seeds due to food security needs and community-based farming systems enable extension services while those that hinder include some cultural preference for "natural seeds" and traditional farming methods and caution toward genetic modification of food crops. **Ethical and religious** factors that advance Biotech & GEd include, a supportive environment for scientific progress; strong government oversight ensures ethical compliance and the existence of biomedical ethics committees while those that hinder include ethical concerns about gene editing in humans/animals and public sensitivity to GM technologies in food.

3.6 Genome Editing Programmes and Projects

Currently, Rwanda has no active GEd projects. However, it has ongoing GMO work focused on priority staple crops, particularly banana, cassava, and Irish potato. The ongoing GMO and planned trials target traits such as disease resistance, pest resistance, drought tolerance, and yield improvement (Table 3). A flagship initiative, the Rwanda Agricultural Biotechnology Programme (approximately US$14 million; December 2025–for five years), is spearheaded by RAB in partnership with CIP (Partner), Danforth Plant Science Centre, MSU (potato), IITA (banana).

Table 3: Biotech/GEd projects and programmes in Rwanda

Projects	Collaborating partners	Biotech/GEd Technique	Stage (Lab, field trial, commercialisation)	Funding (US$)	Funding source	Reference
Rwanda Biotech Capacity Building Rwanda Agricultural Biotechnology Programme- Banana (disease resistance), *Potato (disease resistance), cassava (virus resistance, yield)*	RAB (lead implementation), CIP (Partner), Danforth Plant Science Centre, MSU (potato), IITA (banana)	Transgenic (GMO) mixed with molecular breeding depending on the trait in question	Early: lab & preparation; pilots/field stacking of cassava, programme launched December 2025.	$14 million over five years to strengthen research infrastructure and human capacity.	Bill & Melinda Gates Foundation (grant implemented via CIP/RAB	https://cipotato.org/event/rwanda-biotech-capacity-building-biocap-project-launch/, Rwanda to Build $14M Biotech Centre for Staple Crops - Ecofin Agency, Rwanda Launches $15 Million Global Crop Biotechnology Initiative to Boost Food Security - Empower Africa, https://www.iita.org/news-item/rwanda-launches-biocap-project-with-iita-as-key-biotechnology-partner/,

3.7 Human Capital and Institutional Capacity

3.7.1. Summary of Existing Human Capacity

Rwanda has a small but steadily growing pool of scientists trained in GEd, primarily drawn from national research institutions and public universities, most notably the University of Rwanda. This capacity is supplemented by researchers who have participated in regional short courses, fellowships, and intensive CRISPR training programs offered through international and continental initiatives. While these efforts have strengthened foundational skills, the overall depth of expertise remains limited.

Engagement by private universities in genome editing is uneven, and formally accredited GEd-specific modules are absent from most higher education curricula. As a result, training in genome editing is often embedded within broader molecular biology or biotechnology courses rather than delivered through dedicated programs. This constrains the development of a critical mass of specialists with advanced, hands-on competencies in genome editing applications.

Significant investments are still required to scale up laboratory infrastructure, establish accredited and standardized curricula, and create clear academic and professional career pathways in genome editing. Strengthening institutional capacity will depend not only on infrastructure and equipment, but also on sustained training, staff retention strategies, and incentives that encourage researchers to remain active in national GEd programs.

3.7.1.1. Women, Youth and Special Interest Groups Involvement in Modern Biotech and GEd

Rwanda presents a highly progressive environment for gender inclusion and youth engagement, backed by aggressive top-down policies like the Second National Strategy for Transformation (NST2) running through 2029 and the Gender and Youth Mainstreaming Strategy managed by the Ministry of Agriculture (MINAGRI).

Women and youth in Rwanda are increasingly positioned at the centre of the country's agricultural transformation, driven by strong state-led promotion of science, technology, engineering, and mathematics The structural disparity in biotechnology R&D is being systematically addressed through targeted initiatives, high-tech infrastructure investments, and international capacity-building programs. Women are actively steering critical biotechnology portfolios. For instance, the Plant Micro-Biotechnology Program at the Rwanda Agriculture and Animal Resources Development Board (RAB) at the Rubona research station is heavily led and coordinated by female plant scientists. National initiatives like the annual DNA Week explicitly target young university students and biotechnology undergraduates to build the next generation of gene-editing (GEd) and synthetic biology experts.

Rwanda's macro-policy framework mandates the integration of youth or persons with disabilities into high-tech solutions including GEd laboratories. The second national strategy for transformation (NST2) explicitly targets a 50% increase in national agricultural productivity and mandates creating tech-driven opportunities for youth and women to reduce rural underemployment. The MINAGRI Gender and Youth Mainstreaming Strategy policy aligns directly with the strategic plans for agricultural transformation. This legally ensures that women and youth benefit equally from agritech interventions, ensuring funding and training for female-led cooperatives and young "agri-preneurs" utilizing advanced agricultural tools. The Rwanda Agricultural Biotechnology Programme heavily emphasizes capacity strengthening. It provides the financial and infrastructural runway for young local researchers to master and deploy GEd and genetic transformation to combat regional threats like Cassava Brown Streak Disease and potato late blight.

3.7.2 Research, Development, and Academic Institutions

Several universities and research institutions in Rwanda either possess existing capacity or demonstrate strong potential to support genome editing (GEd) technologies in agriculture and related fields. These institutions exhibit varying strengths and are positioned to play complementary roles in GEd research, training, and implementation.

Key research institutions include the Rwanda Agriculture and Animal Resources Development Board which leads national agricultural research efforts in crop improvement, seed systems, crop protection, livestock breeding, and trait deployment. Rwanda Agriculture and Animal Resources Development Board plays a significant role in integrating genome editing into applied agricultural research and pilot projects.

International research organisations with Rwanda offices, such as the International Institute of Tropical Agriculture (IITA) and the International Potato Centre (CIP), collaborate closely with national institutions. These organisations provide technical support, training workshops, germplasm, and research material, thereby strengthening Rwanda's exposure to modern biotechnology and GEd approaches.

The Rwanda Higher Education Council plays a critical regulatory role by accrediting academic programs, evaluating higher education standards, and ensuring quality assurance. HEC also provides equivalency recognition for degrees obtained outside Rwanda and, in collaboration with government and foreign universities, supports scholarship programs in modern biotechnology for Rwandan scientists.

Overall, institutional capacity varies significantly across organisations. While some institutions possess foundational human capital and infrastructure, others require substantial upgrading. Targeted advanced training, tailored to institutional mandates and existing capacity levels, will be essential to enable Rwanda to build a coordinated and sustainable genome editing research and innovation ecosystem.

3.7.3 Training and Professional Development

Rwanda's capacity in GEd is being strengthened primarily through short-term specialized training programs, regional centres of excellence, and international collaborations (Table 4). These opportunities complement limited in-country postgraduate offerings and play a critical role in building foundational and applied competencies among Rwandese researchers, regulators, and practitioners.

3.7.4 Training Courses on Genome Editing in Rwanda

Regional and continental training platforms

Rwandese scientists regularly access training through the Biosciences eastern and central Africa–International Livestock Research Institute (BecA–ILRI) Hub in Kenya, which offers annual courses in CRISPR/Cas genome editing tools, molecular biology, and bioinformatics. In addition to short courses, BecA–ILRI has hosted Rwandese MSc and PhD students, providing advanced research exposure and hands-on laboratory experience.

The AUDA-NEPAD, through the African Biosafety Network of Expertise (ABNE), provides training focused on biosafety, risk assessment, and policy and regulatory frameworks for biotechnology and genome editing. These programs strengthen institutional and regulatory capacity rather than laboratory skills alone.

The Open Forum on Agricultural Biotechnology in Africa (OFAB) supports advocacy, science communication, and public engagement around biotechnology and GEd. OFAB collaborates with Rwanda Agriculture and Animal Resources Development Board in awareness creation and stakeholder engagement related to biotechnology initiatives.

Crop-focused and applied training initiatives

International research and development organisations, including IITA, CIMMYT, and AATF, provide crop-specific training linked to breeding programs in banana, cassava, and maize. These initiatives combine molecular breeding, genome editing concepts, and applied research exposure within African crop improvement networks.

Programs such as AgShare.Today occasionally offer short courses or workshops in molecular biology and genome editing, typically aligned with regional breeding and research initiatives.

Specialized genome editing courses

Rwanda has also hosted advanced genome editing training locally. Notably, the course “Gene Editing: A Short Course for African Biosciences Professionals” was held in Kigali from 25–29 November 2024 at the University of Rwanda’s Regional Centre of Excellence in Biomedical Engineering and e-Health. The four-day in-person training brought together 39 participants from 20 African countries and provided comprehensive instruction in CRISPR/Cas9 genome editing techniques, with applications in agriculture and public health. Uru Research and Development Group, the University of Dar es Salaam, the Rwanda Biomedical Centre, and the GeneConvene Global Collaborative of the Foundation for the National Institutes of Health (FNIH).

Rwandese researchers also participate in the African Plant Breeding Academy (AfPBA) CRISPR Course, organized by the Innovative Genomics Institute (IGI), African Orphan Crops Consortium (AOCC), Seed Biotechnology Centre (UC Davis), and IITA in Nairobi, Kenya. This program offers intensive, hands-on CRISPR-based genome editing training using banana as a model crop and targets PhD-level plant scientists working in national agricultural research programs.

Overall assessment

While these training opportunities have significantly enhanced exposure and foundational skills in genome editing, most are short-term and externally hosted. Sustained national capacity will require the institutionalisation of GEd training within accredited university curricula, expansion of in-country advanced laboratories, and structured career development pathways to retain trained scientists within Rwanda’s research and innovation ecosystem.

The University of Rwanda, the country’s main public university, is the central academic institution for biotechnology education and research. Its’ College of Agriculture, Animal

Sciences and Veterinary Medicine (CAVM) and College of Science and Technology (CST) have established programs in molecular biology, genetics, and biotechnology, supporting both undergraduate and graduate training. UR remains the only institution in Rwanda currently offering a graduate program in biotechnology, making it a cornerstone for national human capital development in GEd (Table 5).

Carnegie Mellon University Africa contributes specialized expertise in information and communication technologies, data science, artificial intelligence, bioinformatics, and machine learning, which are increasingly relevant to genome editing research, data-driven breeding, and precision agriculture. The Rwanda Institute for Conservation Agriculture (RICA), a U.S.–Rwanda collaborative institution, focuses on conservation agriculture, livestock genetics, sustainability, and applied field trials. RICA's strong practical orientation positions it as an important partner for translating laboratory innovations into field-level applications.

Other universities, including Kigali Independent University (ULK) and the Adventist University of Central Africa (AUCA), offer agriculture-related programs and basic biotechnology or molecular biology courses, though their engagement in advanced genome editing research remains limited. INES-Ruhengeri offers an undergraduate biotechnology program, contributing to foundational skills development.

Table 4: Overview of training programmes on GEd

Institution / Organizer	Training Programme	Target Audience / # of Trainees (annual)	Frequency	Duration	Gaps Identified
University of Rwanda (UR) + Uru Research & Development Group + Rwanda Biomedical Centre + FNIH (GeneConvene)	Gene Editing: Short Course for African Biosciences Professionals (2024)	African bioscience professionals; ~20–30 per cohort	One-off (planned for Nov 2024; potential to repeat)	5 days	Limited to introductory exposure; not yet institutionalized as a recurring national programme; advanced lab-based hands-on training capacity in Rwanda is still minimal.
IITA–CGIAR + IRRI	Genome Editing in Crops Workshop (Nairobi, 2024)	African crop scientists, regulators; ~30–40 participants	Periodic (recently Oct 2024; not annualized)	4 days	Regional, not Rwanda-specific; focuses more on regulatory/policy context than full lab training; limited seats for Rwandans.
ICRISAT + APCoAB/APAARI + BIRAC-BioNEST	Gene Editing Training Workshop (Hyderabad, India, 2019)	Crop scientists from Africa/Asia; ~25–30 trainees	Occasional (not yearly)	2 weeks	Travel-intensive; not easily accessible; no consistent follow-up support for African participants.
African Plant Breeding Academy (AfPBA) – UC Davis, IGI, IITA, AOCC	AfPBA CRISPR Training Course	Doctoral-level African plant scientists; 10–20 per cohort	Every 1–2 years	6 weeks (3 × 2-week modules across one year)	Highly competitive; small cohorts; only a few Rwandan scientists selected per intake; long-term institutional support needed for participants to apply skills at home.
International Collaborations (e.g., ILRI, CGIAR networks)	Workshops on genome editing regulation, bioinformatics, and livestock genetics	Regulators, policymakers, animal/crop	Ad hoc	3–5 days	Often policy-focused rather than technical; limited direct transfer of CRISPR lab skills; gaps in continuity for Rwanda-based trainees.

		scientists; 20–30 per workshop			

Table 5: Universities offering trainings on Biotechnology and GEd related courses in Rwanda

University	**Courses offered**				
	Certificate/ Short Courses	**Diploma**	**BSc/ Bachelor's**	**MSc/ Master's**	**PhD**
University of Rwanda	Biotechnology workshops, genomics, vaccinology, molecular biology trainings	Laboratory sciences diplomas (through affiliated colleges)	BSc Biotechnology, BSc Biochemistry, BSc Biomedical Laboratory Sciences, BSc Molecular Biology, Genetics-related sciences	MSc Biotechnology, MSc Vaccinology, MSc Medical Products Regulatory Affairs, MSc Pharmaceutical Analysis & Quality Assurance	PhD Biotechnology, PhD Biomedical Sciences, PhD Genomics, PhD Molecular Sciences
INES Ruhengeri	Laboratory & biotechnology trainings	Applied science diplomas	BSc Biotechnology (Food Biotechnology option, Plant Biotechnology option), BSc Biomedical Laboratory Sciences	MSc Biomedical Laboratory Sciences, MSc Food Science & Technology	Limited research postgraduate pathways
Rwanda Polytechnic	Technical laboratory short courses, applied bioscience trainings	Diploma in Laboratory Sciences, Advanced Diploma in Applied	Bachelor of Technology programmes related to laboratory technology, applied sciences, food processing, agriculture and	Emerging applied technology	Not yet strongly established in biotechnology

		Sciences, health technologies	industrial biotechnology support areas		
University of Technology and Arts of Byumba (UTAB)	Agriculture and environmental trainings	Agriculture & environmental management diplomas	BSc Agriculture, Environmental Management, Renewable Energy, IT (limited direct biotechnology programmes)	Developing postgraduate applied science pathways	Research collaborations only

3.8 Infrastructure and Equipment

Rwanda's biosafety laboratory infrastructure is evolving but remains limited in both scope and technical capacity to fully support advanced genome editing research and development. The University of Rwanda and RAB host the country's principal molecular biology and biotechnology laboratories, which operate at Containment Level 2**.** These facilities are functional for routine molecular biology, tissue culture, and preliminary biotechnology research (Table 6).

However, existing laboratories lack several components required for advanced GEd. Key gaps include the absence of CRISPR-optimized equipment, limited access to high-throughput genotyping and validation platforms, inadequate containment greenhouses, and the absence of dedicated quality assurance and quality management systems aligned with international biosafety standards. While current facilities can support early-stage GEd experiments, they are not yet equipped to conduct advanced validation studies, confined field trials, or scaled biosafety testing required for regulatory decision-making and commercialisation. Other institutions, including Rwanda Polytechnic**,** INES-Ruhengeri**,** and RICA**,** maintain basic teaching and training laboratories that contribute to foundational skills development. However, these laboratories lack formal biosafety certification, specialized containment infrastructure, and advanced molecular equipment necessary for sensitive GEd research.

Table 4: Status and needs assessment of biosafety laboratory facilities by universities

Institution	Type of Facility	Containment Level	Status	Limitations	Support Needed
University of Rwanda (UR) – College of Science & Technology; Centre for Human Genetics (CHG)	Molecular biology & genetics labs; biomedical genomics	CL-2 (routine molecular genetics)	Functional (basic)	Equipment aging; insufficient containment for genome-editing experiments; limited number of trained biosafety officers.	Modernisation of facilities; training in biosafety and bioethics; expansion to CL-3 for biomedical/genomic research.
Rwanda Polytechnic (RP) & affiliated Technical Colleges	Applied science labs (food tech, agro processing)	CL-1	Functional (basic teaching)	Not designed for genetic engineering or pathogen research; no biosafety	Capacity building in applied biosafety; create linkages with higher-level labs for training.

				containment levels above CL-1.	
University of Technology and Arts of Byumba (UTAB)	Basic life sciences & agriculture labs	CL-1	Functional (introductory)	Limited facilities; lacks biosafety cabinets, molecular tools, or editing capacity.	Investment in core infrastructure (CL-2), biosafety training modules.

Across institutions, several cross-cutting constraints were identified. These include limited biosafety certification and compliance systems, inadequate diagnostic and containment equipment, shortages of trained biosafety and bioethics officers, and insufficient quality management frameworks (Table 7). Collectively, these gaps restrict Rwanda's ability to safely conduct high-risk or high-value GED experiments and to transition promising research outputs from the laboratory to controlled field testing.

Table 7: Status and Needs Assessment of Biosafety Laboratory Facilities by Research Institution

Institution	**Type of Facility**	**Containment Level**	**Status**	**Limitations**	**Support Needed**
Rwanda Agriculture & Animal Resources Development Board (RAB)	Plant & animal biotechnology labs (crop protection, genetic improvement)	CL-2 (plant & animal molecular labs); CL-3 for select veterinary labs	Functional (limited capacity)	Inadequate advanced genome-editing equipment; limited containment facilities for high-risk pathogens; lack of fully accredited CL-3 plant/animal facility.	Upgrade plant biotech labs to CRISPR-capable standards; strengthen molecular diagnostics; establish accredited CL-3 for animal diseases.
Rwanda Biomedical Centre (RBC) / National Reference Laboratory (NRL)	Public health microbiology & molecular labs	CL-2 and some CL-3 functions (TB, emerging pathogens)	Functional (strategic capacity)	Focuses on human health; facilities not readily accessible for agricultural GEd;	Investment in cross-sectoral "One Health" labs; establish genome-editing modules for biomedical & zoonotic pathogens.

Rwanda Institute for Conservation Agriculture (RICA)	Agricultural training & applied research labs	CL-1 / teaching labs	Functional (teaching only)	Lacks containment & molecular biology infrastructure for gene editing or transgenics; designed for applied agronomy.	Development of biotech research wing; partnerships with RAB/UR for biosafety-ready facilities.
Private / Regional Partnerships (e.g., collaborations with AATF, CGIAR, ILRI)	Shared molecular labs (crop biotech, livestock genetics)	Typically, CL-2 in partner hubs	Accessible through partnerships	No permanent in-country access to advanced CRISPR-ready or CL-3 labs; reliance on external partners for complex work.	Build national centres of excellence; create regional hubs (with EAC, AUDA-NEPAD support) for genome-editing R&D

Addressing these limitations will require targeted investments to upgrade selected facilities to CL-2+ or CL-3, where appropriate, alongside the development of specialized containment greenhouses for agricultural trials. Additional priorities include training and accreditation of biosafety and bioethics personnel, acquisition of advanced GEd and validation tools, and establishment of robust institutional quality management systems. Strengthening biosafety infrastructure will not only enhance research capacity but also reinforce public confidence and regulatory trust, creating an enabling environment for the responsible and safe deployment of GEd technologies in Rwanda. Table 7 presents a summary of status and needs assessment of biosafety laboratory facilities across key institutions in Rwanda.

3.9 Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/ Breeds for Improvement Using Genome Editing

Rwanda's traditional and staple crops, livestock, agroforestry species, and fisheries present significant opportunities for improvement through genome editing, with direct implications for food security, climate resilience, and economic growth. Strategic targeting of these organisms aligns with national development priorities and the need to address persistent productivity constraints across agricultural systems. Based on the National Agricultural Policy and stakeholder consultations (Ministry of Agriculture and Animal Resources, 2018), Rwanda's priority organisms for genome editing span crops, livestock, agroforestry, and fisheries, reflecting the need to balance food security, economic growth, and environmental sustainability.

Among crops, cassava, maize, beans, and Irish potato remain the highest priorities due to their contribution to household nutrition and national food availability. Genome editing offers strong potential for traits such as virus resistance in cassava, drought tolerance and pest resistance in maize, and late blight resistance in potato. Existing R&D efforts, primarily through RAB, University of Rwanda, and regional partnerships, provide an entry point for advancing these traits toward application. Cassava, maize, and potato have already attracted early-stage genome editing research through national and regional collaborations, providing a foundation for further advancement.

In the livestock sector, dairy cattle and poultry are prioritized because of their role in food and income security. Target traits include disease resistance, heat tolerance, and improved feed efficiency, which could significantly improve productivity and profitability for smallholder producers. Despite the high potential, genome editing research in livestock remains at an early stage, with most efforts limited to conventional breeding and exploratory studies.

For agroforestry, priority species include Eucalyptus, Grevillea, and selected traditional fodder and multipurpose trees. Genome editing could support faster growth, enhanced pest and disease resistance, and improved adaptability to changing climatic conditions, thereby strengthening integrated farming systems and ecosystem services.

In fisheries, Nile tilapia represents a strategic candidate for genome editing due to its economic importance and expanding aquaculture sector. Target traits include enhanced growth performance and disease resistance, which could reduce production costs and improve competitiveness of domestic fish production.

By prioritizing these organisms, Rwanda can harness genome editing not only to increase productivity but also to reduce vulnerability to climate change, lower production costs for smallholders, and sustain rural livelihoods. Successful adoption, however, will depend on aligning technological interventions with socio-economic priorities, cultural contexts, and robust biosafety oversight.

Current research and Development capacity is strongest in crop biotechnology, driven by national institutions such as RAB and University of Rwanda in collaboration with international partners. In contrast, genome editing research in livestock, fisheries, and agroforestry remains limited, confined to proof-of-concept studies or conventional breeding approaches. Priority Organisms for Genome Editing Application are presented in Table 8 below.

Table 8: Priority staple and traditional crops that could potentially benefit from GE technology

Crops	Trait improved of interest	Socio-Economic Justification	GEd Potential (Low/Medium/High)	Existing R&D	Actual vs Expected Annual Production Capacity (tonnes)
Cassava	Virus (CMD/CBSD) resistance; higher starch/yield; reduced cyanogenic potential; drought tolerance	Staple for food security in marginal areas; supports >700k households; second only to bananas in calorie contribution.	High — well-characterized targets (virus resistance, yield traits).	RAB-led cassava work; IITA/CIAT/IITA breeding networks; FAO/IFAD support programmes; (RAB + AATF) targeting cassava. (BioMed Central, AATF) (Gmakouba etal., 2024)	Actual (2023): ~1.34 million t (annual total, 2023). (FreshPlaza, ReportLinker) (National Institute of Statistics of Rwanda, 2023).
Cooking banana (East African highland / plantain types)	Resistance to Xanthomonas wilt and nematodes; improved yield and shelf life; starch quality	Primary staple (largest single crop by tonnes).	High — many targets (disease resistance) trait delivery into local landraces is feasible but regeneration protocols vary by cultivar.	RAB & partners (IITA, One Acre Fund trials), CGIAR banana programs; breeding & agronomy work ongoing. GEd banana work regionally (IITA/CGIAR) provides training pathways. (One Acre Fund, Wiley Online Library)-(Musabyemungu etal., 2025)	Actual (2023, Season A): ~1,219,408 t (season A). Expected / projected: banana remains the top-volume staple. (NISR)
Irish potato (Solanum tuberosum)	Late blight resistance, improved seed quality (EIG	Major staple and commercial crop. Seed system weaknesses	Medium–High — potato is amenable to GEd (blight	Strong national focus (RAB, UR agronomy research);	Actual (2022/2023): ~900,000 t (reported 2022 national

Crops	Trait improved of interest	Socio-Economic Justification	GEd Potential (Low/Medium/High)	Existing R&D	Actual vs Expected Annual Production Capacity (tonnes)
	seed systems), bacterial wilt resistance, storage life	cause yield gaps and postharvest losses.	resistance, tetraploid genetics complicate editing;	programme. (FAOHome, minagri.gov.rw) (Ministry of Agriculture and Animal Resources, 2021)	production estimates). (FAOHome, PotatoPro)
Maize	Drought tolerance, pest resistance (legume stem borer), improved nutrient use efficiency, quality protein maize (nutritional)	Major cereal for diets and animal feed;	High — maize is a model crop for GEd; multiple successful CRISPR edits.	RAB, CIMMYT, AATF/CMP programme; CGIAR (IITA, CIMMYT) collaborations; ongoing breeding & seed programs. (AATF, IPAD)	Actual (recent annual range): ~480–508k t (2023 estimates/PS&D reporting). (Tridge, FAOHome)
Common beans (Phaseolus vulgaris)	Pest/disease resistance, increased yield stability, reduced cooking time, biofortification (iron, zinc)	Widely planted (~80% households grow beans); key protein & income source; high per-capita consumption	Medium — GEd for improved cooking time and nutrient content is feasible; polygenic yield traits more complex.	National breeding (RAB), CGIAR/CIAT bean programs, value-chain studies (postharvest loss work); (Open Knowledge FAO, CGSpace) (Delgado etal., 2024)	Actual (2022): ~449,000 t (dry beans, FAOSTAT/Helgi). (Helgi Library)
Sweet potato	Virus resistance, beta-carotene (vitamin A) enhancement, storage/quality	Important for nutrition (beta-carotene varieties address vitamin A deficiency)	Medium — GEd for provitamin A and virus resistance is technically feasible;	RAB, CIP (regional), FAO/IFAD projects on root crops. (BioMed Central, IAEA)	Actual (Season A 2023): sweet potato production ~454,355 t (Season A 2023). (NISR)

Source: National Agricultural Policy and the National Institute of Statistics

3.10 Intellectual Property Rights and Benefit Sharing

Rwanda has recently undertaken significant reforms to strengthen its intellectual property and benefit-sharing frameworks. The enactment of Law No. 055/2024 of 20/06/2024 on the Protection of Intellectual Property, which entered into force on 31 July 2024, represents a major overhaul of the national IP regime. The law enhances protection of intellectual property, explicitly covers plant varieties and breeders' rights, and aligns Rwanda's IP system with international standards.

Rwanda's ratification of the Arusha Protocol (effective November 2024) further enables plant breeders to seek protection for new plant varieties across multiple African countries through a single application, thereby facilitating regional commercialisation. In addition, Rwanda's accession to the Budapest Treaty streamlines patenting of microorganisms by recognizing deposits made in international depositories an important provision for modern biotechnology and genome editing (GEd) innovations.

The country's IP system is governed primarily through the Rwanda Development Board (RDB), the Ministry of Agriculture and Animal Resources (MINAGRI), and biosafety authorities under environmental governance frameworks. Rwanda is also a member of major international IP and biosafety treaties including TRIPS, ARIPO, the Cartagena Protocol on Biosafety, and the Patent Cooperation Treaty (PCT) (Table 9).

Together with the 2024 biosafety law, these reforms align Rwanda's national framework with global instruments such as TRIPS, the Cartagena Protocol on Biosafety, the Nagoya Protocol on Access and Benefit Sharing (ABS), and the Nagoya–Kuala Lumpur Supplementary Protocol on Liability and Redress. Institutional responsibilities are clearly delineated: the Rwanda Development Board (RDB) manages intellectual property administration, while REMA oversees biosafety regulation and ABS implementation.

Research institutions such as RAB and the University of Rwanda often working in partnership with AATF, CGIAR centres, and other international collaborators are actively engaged in biotechnology and GEd projects. These collaborations typically involve negotiated licensing and IP arrangements, although most remain governed by project-specific agreements rather than nationally tested commercialisation precedents.

Despite this progress, important gaps remain. Rwanda's ABS system is still being operationalized, with ministerial orders pending to clarify procedures related to access, benefit sharing, and emerging issues such as digital sequence information. Local institutions also have limited in-house capacity for intellectual property portfolio management, license negotiation, and enforcement of benefit-sharing agreements, leaving them heavily dependent on international partners. Without stronger technology transfer offices, clearer commercialisation pathways, and institutional IP management capacity, Rwanda risks under-capturing value from its genetic resources and GEd innovations. Addressing these gaps will be essential to ensure that the strengthened legal framework translates into practical, equitable, and economically beneficial outcomes.

Table 5: A summary of IPRs, their institutions and laws in Rwanda

Intellectual Property Right (IPR)	**Organization / Institution**	**Laws (Acts) / Regulations / Protocols / Treaties**
Patents for Biotechnology & Genome Editing	Rwanda Development Board (RDB); Registrar General	Law N° 055/2024 on the Protection of Intellectual Property; Patent Cooperation Treaty (PCT); ARIPO Harare Protocol; TRIPS Agreement (WIPO)
Plant Breeders' Rights (PBRs)	MINAGRI; Rwanda Inspectorate, Competition and Consumer Protection Authority (RICA); RDB	Law N°005/2016 Governing Seeds and Plant Varieties; Law N° 055/2024 on the Protection of Intellectual Property; UPOV-aligned principles (ECOLEX)
Protection of Plant Varieties & Seed Certification	MINAGRI; RICA	Law N°005/2016 Governing Seeds and Plant Varieties (ECOLEX)
Biosafety & Living Modified Organisms (LMOs)	Rwanda Environment Management Authority (REMA); National Biosafety Committee	Law N° 025/2024 Governing Biosafety; Cartagena Protocol on Biosafety (UNEP Law Platform)
Utility Models & Agricultural Innovations	Rwanda Development Board	Law N° 055/2024 on the Protection of Intellectual Property (WIPO)
Trademarks for Seed & Biotech Products	Rwanda Development Board	Law N° 055/2024 on the Protection of Intellectual Property; Madrid Protocol (WIPO)
Geographical Indications (GI)	Rwanda Development Board	Law N° 055/2024 on the Protection of Intellectual Property; TRIPS Agreement (WIPO)
Copyright for Research Databases & Biotech Outputs	Rwanda Development Board	Law N° 055/2024 on the Protection of Intellectual Property (WIPO)
Protection Against Unfair Competition	Rwanda Development Board	Law N° 055/2024 on the Protection of Intellectual Property (WIPO)

3.11 Private Sector Participation

The private sector plays a significant role in Rwanda's seed system and will be critical to the eventual commercialisation and scaling of genome-edited products. At present, no LMOs are under commercial release in Rwanda. However, private sector engagement is emerging through public–private partnerships (PPPs) in biotechnology research and seed multiplication. Bayer, in collaboration with RAB and other partners, is participating in the Rwanda Agricultural Biotechnology Programme, which is evaluating biotechnology-derived maize, cassava, and Irish potato for the first time in the country. Improved seed distribution in Rwanda is already well established through partnerships with more than 30 private seed companies, including One Acre Fund–Tubura, Western Seed Company Ltd, Rumbuka Seed Ltd, Export Trading Group, Holland Greentech Rwanda, Tri-Seed Co. Ltd, among others. These companies distribute seed through extensive agro-dealer networks that reach smallholder farmers nationwide (Table 10).

Private seed companies such as Seed Co Rwanda, Holland Greentech, and Seed Effect Rwanda already play a pivotal role in multiplying and disseminating improved varieties, often in collaboration with RAB, CIMMYT, and Alliance for a Green Revolution in Africa. Their established logistics and farmer outreach systems position them as key conduits for future GEd-derived varieties once regulatory approvals are in place.

Public–private partnerships offer promising models for integrating genome editing into Rwanda's agricultural innovation system. For example, RAB's collaboration with AATF on biotech maize and cassava anticipates private sector involvement in large-scale seed multiplication and commercialisation following regulatory clearance. Similarly, partnerships with CGIAR centres such as CIMMYT and CIP facilitate local testing and adaptation of international innovations prior to licensing to private distributors.

Nonetheless, private sector participation in GEd remains limited by gaps in technical expertise, regulatory familiarity, intellectual property management capacity, and access to risk-tolerant finance. Inclusive business models that ensure affordability and accessibility for smallholder farmers are also underdeveloped (Table 11). Strengthening PPP frameworks, clarifying intellectual property and licensing pathways, and providing targeted incentives for local agribusinesses to invest in biotechnology will be essential for translating Rwanda's GEd research potential into tangible productivity and food security gains.

Table 10: Overview of genome editing stakeholders and activities in the private sector

Company/Entity	Type (Agri-biotech, Start-up, etc.)	GEd Activities	Partnerships	Challenges Faced	Investment Interest
Rwanda Seed Co. / Seed Co Rwanda (partners in seed systems)	Seed company / agri-biotech	Seed multiplication, hybrid seed distribution (maize, cereals). Indirect role in GEd via improved varieties pipeline	RAB (Rwanda Agriculture Board), CIMMYT, AGRA, government seed programs, One Acre Fund (Tubura), CIMMYT	Limited local R&D capacity, dependence on imported germplasm, regulatory delays for GM/GEd crops	Moderate – interested in climate-smart seed technologies
Holland Greentech Rwanda	Agri-tech company	Provides greenhouse technologies, high-quality horticultural seeds, supports precision agriculture (enabling platform for future GEd crops)	RAB, horticulture exporters, farmer cooperatives	High input costs, limited biotech R&D integration	High interest in advanced breeding + biotech seeds
One Acre Fund (Tubura)	Agri-tech / social enterprise	Large-scale distribution of improved seeds and fertilizer; supports adoption of biotech-enabled crops	RAB, donors, seed companies, AGRA	Scalability constraints, farmer affordability issues	High interest in climate-resilient biotech seeds
Virunga Biotech	Agri-biotech / commercial farming enterprise	Uses precision agriculture, field trials, crop improvement for export horticulture; supports improved	IDH (Sustainable Trade Initiative), SAIP, export buyers in EU	Limited access to advanced biotechnology tools (CRISPR)	Moderate–High (export-driven biotech agriculture)

		plant varieties (pre-GEd enabling environment) (Virunga Biotech)			
Export Trading Group (ETG Rwanda)	Agribusiness	Supports agro-input supply chain, improved seed distribution and commodity aggregation (enables biotech crop uptake)	RAB, regional markets, farmers cooperatives	Market volatility, weak local biotech integration	Medium
Seed of Trust Ltd & local seed SMEs	Seed SMEs	Seed multiplication and distribution; indirect role in biotech diffusion	Government seed system	Limited research capability, reliance on imported genetics	Low–Medium

Table 6: Challenges and opportunities of engaging in the private sectors

Challenges	Opportunities
Limited advanced genome-editing infrastructure — few private CRISPR/genomics labs	Strong government support for agricultural innovation through PSTA4, NST1, and BIOCAP
High cost of biotech R&D equipment (sequencers, tissue culture labs, gene-editing tools)	Growing investment in biotechnology capacity building by government and donors
Dependence on donor-funded biotech projects	Large international support ecosystem (CGIAR, Gates Foundation, AATF, EU, IITA, CIP)
Shortage of highly specialized biotech professionals (CRISPR, genomics, bioinformatics)	Expanding biotechnology training programs at University of Rwanda and partner institutions
Limited private-sector biotechnology companies	First-mover advantage for biotech startups entering the market
Weak commercialization pipeline from research → market	Strong public research institutions (RAB) can provide technology transfer opportunities
Regulatory uncertainty for genome-edited crops	Existing biosafety and seed systems provide foundation for future GEd regulation
Slow approval process for GM/GEd field trials	Opportunity to shape progressive genome-editing policy frameworks
Limited venture capital for deep-tech biotech startups	Rwanda's innovation-friendly business environment attracts impact investors
Public confusion between GMOs and genome editing	Opportunity for science communication and biotech awareness programs
Low public awareness of biotechnology benefits	Food security concerns create openness to improved crop technologies
Dependence on imported biotech equipment and reagents	Opportunity for local biotech supply chains and laboratory services
Limited intellectual property commercialization culture	Potential for local biotech patent development and licensing systems
Limited industrial biotechnology manufacturing	Potential for agro-processing and biotech manufacturing growth
Limited local private CRISPR intellectual property development	Potential to build regional leadership in agricultural genome editing

3.12 Funding and Investment Landscape

Funding for genome editing in Rwanda is currently dominated by international donors, regional initiatives, and multilateral partners, with limited domestic budget allocations specifically dedicated to advanced biotechnology. Key contributors include the Bill Gates Foundation, USAID, and African Union–linked biosciences initiatives, which support crop improvement programs targeting maize, cassava, and beans through partnerships with RAB and University of Rwanda (Table 12).

Global research consortia, particularly CGIAR centres such as CIMMYT, CIP, and IITA, Danforth Plant Science Centre channel external resources and expertise into Rwanda to evaluate and adapt genome editing applications. These investments primarily support collaborative research, capacity building, and regulatory readiness, rather than late-stage development or commercialisation.

At the national level, funding contributions remain modest and are often embedded within broader agricultural transformation or innovation programs. Institutions such as RAB and UR have accessed competitive grants through the National Council for Science and Technology, but these resources are insufficient to sustain end-to-end genome editing pipelines. The absence of dedicated domestic financing constrains Rwanda's leverage in IP negotiations, limits private sector participation, and increases reliance on externally driven research agendas.

While international funding has been instrumental in building foundational capacity, over-dependence on external resources poses risks to long-term sustainability and national ownership. Expanding domestic co-financing mechanisms, establishing dedicated biotechnology funding windows, and incentivizing private sector investment will be critical to aligning genome editing research with Rwanda's strategic agricultural priorities and ensuring durable socio-economic returns.

Table 12: Overview of national and other funding sources for genome editing

Funder/Donor	Organisation Type	GEd Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
Bill & Melinda Gates Foundation	Philanthropic Foundation	BIOCAP **Project** Rwanda Agricultural Biotechnology Programme-	$14 million- to strengthen research infrastructure and human capacity.	5 years	RAB (lead implementation), CIP (Partner), Danforth Plant Science Centre, MSU (potato), IITA (banana)	CRISPR and other modern biotech tools; Banana (disease resistance), *Potato (disease resistance), cassava (virus resistance, yield)*
Bill & Melinda Gates Foundation	Philanthropic Foundation	Rwanda Agricultural Biotechnology Programme (cassava, maize, potato—disease/pest/drought tolerance)	~$9.9 million	5 years (Oct 2024 – Oct 2029)	AATF as grantee; implementation by RAB	Genome-editing for staple crop resilience and productivity
USAID (via AATF, TELA project)	International Development Agency	Drought- and pest-tolerant maize project (TELA maize)	(Included within BMGF support; not separately quantified)	Not specified (ongoing)	AATF, then implementation in partner NARs such as RAB once Rwanda joined post-biosafety	Maize for insect resistance and drought
African Development Bank (AfDB)	Continental Bank	Support to establish African Pharmaceutical Technology Foundation (APTF)	$11.96 million (plus $1.93M govt of Rwanda co-finance)	Not specified, rollout began Mar 2024	APTF (hosted in Kigali)	Building pharmaceutical R&D & regulatory capabilities

4. RECOMMENDATIONS AND POLICY OPTIONS

Genome editing presents Rwanda with a transformative opportunity to enhance agricultural productivity, strengthen resilience to climate change, and improve national food security. To fully realize these benefits, a coordinated approach is required that integrates clear regulatory frameworks, strengthened institutional capacity, upgraded infrastructure, sustainable financing, and active engagement of both public and private sector actors.

Regulatory frameworks

Rwanda should finalize and operationalize biosafety and intellectual property regulations that explicitly address genome-edited products, ensuring clear, predictable, and science-based approval pathways from laboratory research to field trials and commercialisation. Regulatory clarity will reduce uncertainty for researchers and investors while safeguarding environmental and human health. Continued alignment with regional and international frameworks, including AUDA-NEPAD guidelines, COMESA biosafety initiatives, and global treaties such as the Cartagena Protocol and the Nagoya Protocol will enhance regulatory credibility, promote harmonisation, and facilitate cross-border trade and collaboration.

Capacity building

Sustained investment in human capital is essential for the responsible development and regulation of GEd. National universities and research institutions should integrate genome editing, biosafety, and bioethics modules into undergraduate and postgraduate curricula, while targeted training programs should be expanded for regulators, inspectors, and biosafety officers. Strategic partnerships with international universities, CGIAR centres, and regional biotechnology hubs can accelerate skills transfer and mentorship. In parallel, intellectual property management and technology transfer offices within public research institutions should be strengthened to support licensing, commercialisation, and equitable benefit-sharing arrangements.

Infrastructure and equipment

Upgrading research infrastructure is critical to enabling safe and effective GEd research. Existing laboratories at the University of Rwanda (UR) and the Rwanda Agriculture and Animal Resources Development Board (RAB) should be enhanced to meet Containment Level 2+/3 standards, allowing for advanced genome editing and containment of regulated organisms. Investment in controlled greenhouse facilities, confined field trial sites, and aquaculture research systems will support the transition from laboratory research to applied testing. Where feasible, Rwanda should also leverage regional shared infrastructure to reduce costs and accelerate innovation.

Strategic funding and investment

Rwanda should establish a dedicated national innovation fund for genome editing, focused on priority challenges such as cassava disease resistance, climate-resilient maize, improved dairy productivity, and enhanced tilapia production. Blended financing models—combining government allocations, international donor support, and private sector investment—can improve sustainability and reduce reliance on external funding. Fiscal incentives, including tax relief and co-investment schemes, would further encourage private sector participation and domestic innovation.

Private sector participation

The private sector will be central to scaling GEd innovations from research to farmer adoption. Strengthening public–private partnerships (PPPs) with seed companies, agribusinesses, and input suppliers will be essential for large-scale multiplication, distribution, and market uptake of GEd-derived products. Support should be provided to enhance private sector technical capacity, regulatory compliance, and understanding of IP and licensing requirements, while ensuring that business models promote affordability and access for smallholder farmers.

Networking and public engagement

Rwanda should deepen engagement with African biotechnology platforms, regional biosafety networks, and CGIAR research systems, while establishing a national genome editing consortium to coordinate stakeholders across government, academia, industry, and civil society. Equally important is proactive public engagement. Structured consultations, awareness campaigns, and transparent communication with farmers, consumers, and community leaders will be critical for building trust, addressing ethical and cultural concerns, and fostering informed, inclusive adoption of genome editing technologies.

Conclusion

Rwanda is well positioned to leverage genome editing as a catalyst for agricultural transformation. By strengthening regulatory systems, investing in capacity and infrastructure, mobilizing sustainable financing, and fostering inclusive public–private partnerships, genome editing can contribute significantly to national food security, economic growth, and long-term resilience to climate and production shocks.

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ANNEXES

Annex 1: Criteria for Determining Laboratory Status for CL-1 and CL-2 Operations

(i.) **Criteria for infrastructure and equipment for CL 1:** 3-4 rooms containing the following: PCR, Incubator, Sequencers, Freezers (-80, -20), P/ATC room, Access to consumables, LAF chamber, Electrophoresis Apparatus, Autoclave, Microwave, Vortexer, UV illuminator

(ii.) **Criteria for infrastructure and equipment for CL 2:** Standard Microbial Practices + Special practices + All CL-1 equipment plus a mandatory biosafety hazard sign, special protective gear, special Cabinets (class II), controlled access to rooms etc., handling agents of moderate potential hazards to people + animals + environment

	Conditions	**Status**
CL 1	If all in (i) above are available with or without the sequencer	Fully equipped
	Missing any of the other equipment in addition to the sequencer	Not fully equipped
CL 2	Conformance to the criteria in (ii) above	Fully equipped
	Any non-conformance to the criteria in (ii) above	Not fully equipped

Annex 2. List of Institutions and Resource Persons Involved in Biotechnology and Genome Editing

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